

National University of Computer and Emerging Sciences
Lahore Campus

Computer Networks (CS3001)

Date: April 6th 2026

Course Instructor(s)

Ms. Sarah Khaleel, Mr. Nauman Moazzam Hayat

Sessional-II Exam

Total Time: 1 Hours

Total Marks: 35

Total Questions: 5

Roll No

Section

Student Signature

-
- Instruction/Notes:**
- Attempt all questions on the separately provided answer sheet (except Q#5).
 - Q#5 is required to be answered on this question paper as per instructions provided in the question statement.
 - Ensure that question paper is attached to the provided answer sheet.
 - In case you have used rough sheets, they should **NOT** be attached to the answer sheet.
-

CLO 1 (Q#1): Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.

Q1: Answer the following MCQs by writing the correct option(s) (A, B, C, D) in the answer sheet: (Any erasing, cutting, overwriting etc. in the answer sheet will not be marked.) [1 * 5 = 5 Marks]

i) A TCP Tahoe sender has: cwnd = 12 MSS, ssthresh = 16 MSS. A packet loss is detected via 3 duplicate ACKs. What will be the new cwnd immediately after this event?

- A. 6 MSS B. 1 MSS C. 8 MSS D. 11 MSS

ii) A Go-Back-N protocol uses: Window size = 4, Sender transmits packets: 0,1,2,3. Packet 1 is lost. Packets 0, 2 and 3 arrive at receiver. How many packets will be retransmitted?

- A. 1 B. 2 C. 3 D. 4

iii) A TCP sender has: cwnd = 14 MSS, rwnd = 10 MSS. What is the maximum number of unacknowledged segments that can be sent?

- A. 4 B. 10 C. 14 D. 24

iv) The service model of the internet's network layer (IPv4) can be best described as:

- A. Unreliable B. Fast C. Unreliable & Fast D. Best Effort

v) A host obtains an IP address for itself via:

- A. DHCP B. DNS C. HTTP D. FTP

CLO 2 (Q#2,3 and 4): Demonstrate the basics of network concepts using state-of-the-art network tools/techniques.

Q2: UDP, TCP & IP use 1s complement for their checksums. Suppose you have the following three 8-bit words: 01010011, 01100110, 01110100. What is the 1s complement of the sum of these three 8-bit words? (Note that although UDP, TCP or IP use 16-bit words in computing the checksum, for this problem you are being asked to consider 8-bit words instead of 16-bit words.) [5 Marks]

National University of Computer and Emerging Sciences

Lahore Campus

Q3: A TCP sender uses the following parameters:

[2 * 3 = 6 Marks]

- Initial EstimatedRTT = 200 ms, DevRTT = 20 ms
- $\alpha = 0.125$, $\beta = 0.25$
- New SampleRTT = 260 ms

- (a) Compute the new EstimatedRTT
- (b) Compute the new DevRTT
- (c) Compute the new TimeoutInterval

Q4(a): A cloud provider has four separate clusters in a data center. Find the most specific summary route (in dotted decimal CIDR notation including prefix) that includes all four of these subnets: **[5 Marks]**

Subnet 1: 172.16.12.0/24

Subnet 2: 172.16.13.0/24

Subnet 3: 172.16.14.0/24

Subnet 4: 172.16.15.0/24

Q4(b): A router receives a packet with the destination IP 172.16.14.200. The router's forwarding table is:

Prefix Entry	Output Interface
172.16.0.0/16	0
172.16.14.0/23	1
172.16.14.128/25	2
172.16.14.192/26	3

On which interface will the packet be forwarded to?

[4 Marks]

CLO 3 (Q#5): Demonstrate various classical routing and switching protocols via simulations.

Q5: Answer to this question is to be provided by completing the table given below **on the question paper**.

A student brings a brand-new laptop into a university lab. The laptop is configured for dynamic IP addressing. The lab's DHCP server has the IP address 10.1.1.254 and is sitting on the 10.1.1.0/24 subnet. Please fill in the below table with the missing entries for the DHCP messages exchange between the laptop & the DHCP Server in the correct order. **[0.5 * 20 = 10 Marks]**

Step	DHCP Message Name	Source IP Address	Source Port Number	Destination IP Address	Destination Port Number
1					
2					
3					
4					